Outcome: Asthma diagnosis was defined as meeting one or more of the following criteria from age 4 to 6 years (24-month time frame):

- ≥1 inpatient/23hr/ED OR ≥2 physician office visits (separate by at least 30 days) with an ICD-9 code of 493.xx (asthma) or ICD-10 code J45.xx (asthma)
- ≥2 dispensing events with controller medication OR ≥1 dispensing events with controller medication and ≥1 dispensing events with rescue medication
- ≥2 dispensing events with rescue medication

We should include OCS as rescue medications as well.

I think we do not need to include other intravenous rescue medications, as the TennCare dataset does not capture medications administered in the hospital setting (e.g., in the ED or during hospitalization).

<My approach>

Medication classification approach

Pingsheng and her colleagues previously developed a comprehensive asthma medication list based on prior literature and clinical expertise. I have further expanded and refined the medication list using FDA and RxNorm APIs, complemented by manual clinical review. This process resulted in the addition of approximately 5,000 NDC codes, substantially improving coverage and completeness of asthma-related medication identification.

While the original medication list also included drugs relevant to allergic rhinitis, our analysis focuses exclusively on **asthma-related medications**, as allergic rhinitis is not an outcome in the EV study.

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1. Initial enhancement of medication database

We expanded and refined the existing dataset ("*Asthma and AR NDCs_4.29.2025.xlsx*") by integrating information from the **FDA NDC archive and RxNorm APIs**. Then, I manually reviewed all medications in the list. I used the FDA NDC archives rather than the current NDC Directory because the initial list included many obsolete medications. The purpose of this step 1 was to support manual review and improve the classification of asthma medications and their subcategories. In later steps, we will supplement the list using the current FDA NDC Directory.

This step enabled:

- Improved clinical interpretability of each NDC
- Assignment of clinically meaningful medication subclasses
- More transparent review of asthma relevance by clinicians

We defined the following medication subcategories:

- **Rescue medications:** inhaler/oral/intravenous short-acting β2-agonists (SABA), ~~oral corticosteroids (OCS)~~ , short-acting muscarinic antagonist (SAMA), SABA/SAMA, epinephrine inhalers, SABA/ICS
- **Controller medications:** inhaled corticosteroids (ICS), long-acting β2-agonists (LABA), long-acting muscarinic antagonists (LAMA), their fixed-dose combinations, oral methylxanthines, inhaled mast cell stabilizers, non-montelukast leukotriene receptor antagonists (LTRA), 5-lipoxygenase (5-LOX) inhibitors, and biologics
- **Montelukast-specific category:** isolated due to its approval for allergic rhinitis after 2003

Formulation-based subclasses included:
SABA, SABA_oral, SABA_iv, epinephrine_inhaler, SAMA, SABA/SAMA, SABA/ICS, OCS, methylxanthine_iv, ICS, LABA, LAMA, ICS/LABA, LABA/LAMA, ICS/LABA/LAMA, methylxanthine_oral, mast_cell_stabilizers, Non_montelukast_LTRA, 5_LOX, biologics, and Montelukast.

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2. Cross-validation and adjudication

We cross-validated the refined dataset against Pingsheng's curated file ("*Asthma NDCs_5.8.2026.xlsx*").

Discrepancies were resolved through:

- Manual review
- Targeted Google searches for ambiguous NDC products
- Clinical adjudication when classification uncertainty remained

Key decision rules included:

- Retaining Pingsheng's original classification when NDC cannot be identified through APIs (highlighted rows preserved)
- Leaving the classification column blank when rescue vs. controller status could not be reliably determined
- Reclassifying misclassified medications based on route and indication

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3. Key updates

- The following medications/categories were re-categorized as needed:
  - Nasal steroid sprays: not an asthma medication
  - Topical steroid: not an asthma medication

- Oral corticosteroid: not ICS
- 300 ACTUAT isoproterenol hydrochloride 0.16 MG/ACTUAT / phenylephrine bitartrate 0.24 MG/ACTUAT Metered Dose Inhaler [Duo-Medihaler]: combination of a non-selective beta-adrenergic agonist (stimulating both β1 and β2 receptors) and phenylephrine, an alpha-agonist decongestant -> SABA
- Zileuton: non-LTRA leukotriene synthesis inhibitor / 5-LO inhibitor -> 5_LOX
- Nasal LTRA: not an asthma medication
- Cromolyn: oral formulations = non-asthma medication; inhalation = asthma medication
- Theophylline and aminophillin: oral formulations = maintenance therapy; IV formulations = rescue therapy
- ephedrine 2.4 MG/ML / guaifenesin 10 MG/ML / phenobarbital 0.8 MG/ML / theophylline 3 MG/ML Oral Solution: possible maintenance therapy because it contains oral theophylline?
- dyphylline 200 MG / guaifenesin 200 MG Oral Tablet: asthma medication?
- Bulk powder ingredient: Budesonide 1 kg/1 kg POWDER — pharmaceutical ingredient used by compounding pharmacies to create topical creams, ointments, or lotions -> Non-asthma medications because it is not easy to differentiate inhalers and other routes.
- LAMA mono-therapy: e.g., FORMOTEROL FUMARATE. It is not recommended, but was used in the past -> asthma controllers

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4. Expansion of NDC coverage

To improve completeness of the medication list, we expanded NDC coverage using two complementary approaches:

4.1 ATC-based mapping

- Leveraged the **ATC R03 (drugs for obstructive airway diseases)** class to identify additional candidate medications
- Recognized that ATC–NDC mapping is imperfect but useful for broad coverage expansion

  - Downloaded the ATC classification (R03: drugs for obstructive airway diseases; https://atcddd.fhi.no/atc_ddd_index/?code=R03AC&showdescription=yes) and converted ATC codes to NDC codes using the n2c package (https://github.com/fabkury/n2c).
  - From the package file, I linked NDCPACKAGECODE to PRODUCTID.
  - Using PRODUCTID, I extracted inhaler products from the product file.
  - I additionally selected medications with ROUTENAME containing "INHALATION."
  - Finally, I manually reviewed the resulting medication list and confirmed whether each drug was an asthma medication; no items were excluded at this stage.

4.2 NDC structural inference

- Generated additional candidate NDCs using:
  - Labeler and product code patterns (when identifiable)
  - Variations in packaging formats
- This approach cannot perfectly reconstruct labeler and product codes from full 11-digit NDCs, but it enables identification of additional package-code variations.
- I have manually reviewed the additional NDC codes identified to exclude non-asthma medications.

12345	—	6789	—	1

Labeler Code – Product Code – Package Code

Format variations: 4-4-2, 5-3-2, 5-4-1, etc.

Solu-Medrol

Originally contained	00009	—	0039	—	30
Additionally contained	00009	—	0039	—	06